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Fractional Forge — Project a_compact_commercial_airsource_heat_pump

Generated: 2026-05-05

Economics Dashboard

VERDICT: PENDING

EST. UNIT COST

Not computed

[D] Algorithmic Estimate

TARGET COST CEILING

£4,500.00

[A] Brief Constraint

NON-RECURRING ENG. (NRE)

Not computed

[D] Modelled

TOTAL ADDRESSABLE MARKET

Not computed

[B] Research

SUBSYSTEM MODULES

10

[D] Decomposition

BOM LINE ITEMS

0

[D] Parts Generated

MARKET CAGR

Not computed

[C] Analyst consensus

SPATIAL ALLOCATION

Infeasible

[D] Solver verification

Source Grading Key

- [A] Primary datasheet or direct constraint
- [B] Supplier quote or high-confidence research
- [C] Trade journal or secondary source
- [D] Algorithmic estimate or LLM extraction
- [E] LLM assumption or unverified hypothesis

Feasibility Gate Assessment

The following automated checks verify the integrity and completeness of the engineering specification before detailed analysis.

Check	Status	Reason	Evidence
1. Market TAM/SAM/SOM Defined	FAIL	Missing market data	Not computed
2. Regulatory Standards Identified	PASS	Standards found	6 standards
3. Cost Ceiling Specified	PASS	Target provided	£4,500.00
4. Spatial Envelope Provided	PASS	Volume established	75 m³
5. Modules Decomposed	PASS	System architecture built	10 modules
6. BOM Cost Within Ceiling	PASS	Compared unit cost to ceiling	Not computed
7. Risk Matrix Saturated	PASS	FMEA rows generated	Risks exist

Methodology Note

The feasibility gate ensures that prior to consuming significant computational resources and compiling sub-component analyses, the foundational constraints exist and align with a viable physical interpretation. Failure on any PASS/FAIL criteria halts downstream generation.

Data Sources

[System] Internal gate checks

Overall Section Grade: A

1. Brief & Requirements ^[A]

1.1 Mission & Use Case

Mission

To deliver a cost-competitive, R290-propane hydronic split air-source heat pump producing 30kW thermal output at COP e 3.5, purpose-built for small UK commercial buildings and fully MCS certified.

Use Case

The outdoor unit houses the R290 compressor, air-source heat exchanger, and refrigerant-to-water plate heat exchanger; the indoor hydraulic module contains the circulating pump, expansion vessel, controls, and buffer vessel connections. The system provides primary space heating and domestic hot water pre-heat via a low-temperature hydronic circuit at 35–55°C flow temperatures, serving fan coil units, underfloor heating, or low-temperature radiators in small commercial buildings up to approximately 300m² floor area.

1.2 Market Context & Timing ^[B]

UK net-zero legislation (Climate Change Act targets), imminent phase-down of high-GWP refrigerants under revised UK F-Gas Regulation, Boiler Upgrade Scheme commercial expansion, Future Buildings Standard consultation requiring low-carbon heat, and Minimum Energy Efficiency Standards (MEES) enforcement for commercial properties all create immediate demand pull. Gas boiler new-installation restrictions in new commercial builds are anticipated post-2025.

Small commercial building owners and operators (offices, GP surgeries, dental clinics, small hotels, cafes, light industrial units); M&E contractors and installers; property developers targeting EPC-B+ ratings; commercial landlords under MEES pressure; local authorities with small community buildings

Data Sources

[User Input] Original brief provided

[LLM Output] Research synthesis based on prompt

Overall Section Grade: C

1.3 Competitor Landscape [C]

Mitsubishi Electric	
Product	Ecodan CAHV-R290 Commercial / Ecodan QUHZ (R290 series, 14–23kW residential scaling to commercial stacks)
Pricing	Approximately £8,000–£14,000 installed for equivalent 30kW capacity (multi-unit configuration); ex-works equipment cost estimated £5,500–£9,000
Tech Specs	R290 refrigerant; output range 8.5–23kW per unit (multi-unit cascading to 30kW+); COP up to 4.0 at A7/W35; SCOP up to 4.4 (average climate); hydronic split architecture; flow temperatures to 60°C; ErP A+++ rated; sound pressure ~52dB(A)
STRENGTHS Established brand trust with M&E specifiers and installers; extensive UK service network; MCS certified; proven R290 units in market; strong controls integration (MELCloud); high COP performance WEAKNESSES Units in the 30kW single-unit commercial range not yet fully developed in R290; multi-unit cascading adds complexity and cost; premium pricing reduces ROI for cost-sensitive SME customers; primarily residential heritage limits commercial specification credibility	
DIFFERENTIATION ANGLE Target product offers purpose-built single 30kW commercial unit vs cascaded residential units, simpler installation, lower total installed cost, and direct MCS commercial pathway without multi-unit complexity	

Vaillant

Product	aroTHERM plus VWL 125/5 AS (R290, 12.5kW) scaled / aroVAIR commercial range
Pricing	aroTHERM plus 15kW unit approximately £4,200–£5,500 ex-works; commercial aroVAIR equivalents £7,000–£12,000 ex-works; installed costs add 50–100%
Tech Specs	R290 refrigerant on aroTHERM plus residential range up to 15kW; aroVAIR commercial range uses R410A/R32 at 20–100kW; COP 5.0+ at A7/W35 on residential R290 units; hydronic split; ErP A+++ residential; commercial units less transparent on R290 transition timeline

STRENGTHS

Strong European brand recognition; excellent residential R290 performance credentials; broad installer network in UK; robust after-sales service; strong ErP/Ecodesign compliance documentation

WEAKNESSES

Commercial R290 product gap at 25–50kW range; aroVAIR commercial line still predominantly high-GWP refrigerants; pricing positions above target product cost point; commercial sales team less developed than residential channel in UK

DIFFERENTIATION ANGLE

Target product fills the 30kW commercial R290 gap that Vaillant has not yet addressed, with superior commercial-grade hydraulic module and lower ex-works price at volume

Thermia (part of Danfoss Climate Solutions)

Product	Thermia Mega / Danfoss DHP-AQ Commercial ASHP
Pricing	Equipment cost approximately £9,000–£18,000 for 30kW capacity class; installed costs significantly higher due to commercial complexity and limited UK installer base
Tech Specs	R410A/R32 primary refrigerants; 15–60kW thermal output range; COP 3.2–3.8 at A2/W35 depending on model; hydronic architecture; modular outdoor units; flow temperatures to 65°C; DNV/TÜV certified; primarily Northern European market

STRENGTHS

Genuine commercial-grade engineering; high flow temperature capability; strong performance in cold climates; Danfoss compressor and controls integration; long product life expectancy; suitable for retrofit to existing high-temperature systems

WEAKNESSES

No R290 product in commercial range; high-GWP refrigerant exposure to future F-Gas restrictions; premium price point; limited UK distribution and installer network compared to Mitsubishi/Vaillant; less consumer brand recognition in UK SME market

DIFFERENTIATION ANGLE

Target product competes on R290 future-proofing, UK-market MCS certification, lower cost, and stronger UK distribution channel development vs Thermia's Nordic-optimised product and pricing

Daikin

Product	Daikin Altherma 3 R F (EBLQ-D3V3 series) / Daikin Applied Commercial ASHP
Pricing	Altherma 3 F (16–24kW) approximately £4,500–£7,000 ex-works; Applied commercial products £10,000–£20,000+ for 30kW single-unit class; installed costs vary significantly
Tech Specs	R32 refrigerant (Altherma 3 F range, 16–24kW); R410A on larger commercial applied products; COP 4.31 at A7/W35 for 16kW model; SCOP up to 5.07 (average climate); hydronic split; flow to 70°C high-temperature variant; ErP A+++ on residential range; 30kW commercial single unit requires Daikin Applied range with different specification approach

STRENGTHS

Market-leading brand globally; extensive UK installer and service network; broad product range from residential to large commercial; strong technical documentation; MCS certified on residential/light commercial range; good controls ecosystem (Daikin Online Controller)

WEAKNESSES

R32 refrigerant (GWP=675) not future-proof vs R290 (GWP=3); no single 30kW R290 commercial hydronic split product as of 2024; commercial Applied range pricing significantly exceeds target; complexity of commercial products requires specialist commissioning; residential Altherma maxes out below 30kW for single units

DIFFERENTIATION ANGLE

Target product offers the natural refrigerant (R290, GWP=3) advantage Daikin's current 30kW class lacks, at a lower installed cost with purpose-built commercial hydronic module, directly addressing the regulatory trajectory and customer sustainability requirements

Data Sources

[LLM Search] Aggregated competitor data via LLM

Overall Section Grade: D

1.4 Constraints & Targets ^[A]

Target Material	Not computed
Target Process	Not computed
Tolerance Target	Not computed
Quantity Target	Not computed
Batch Size	500
Target Cost Ceiling	£4,500.00
Max Mass	Not computed

Research Sources

Title	Type	Year	Relevance
MCS 007 – Heat Pump Standard for MCS Certification	regulatory_standard	2026	Defines the certification requirements, test conditions, and documentation needed for heat pumps to qualify for the UK Boiler Upgrade Scheme and other government incentives; directly applicable to product certification roadmap
BS EN 14511:2023 – Air Conditioners, Liquid Chilling Packages and Heat Pumps for Space Heating and Cooling	technical_standard	2026	Primary performance test standard defining COP measurement methodology at A2/W35 and other rating conditions used in MCS and Ecodesign compliance testing
IEC 60335-2-40:2022 – Household and Similar Electrical Appliances – Safety – Heat Pumps	technical_standard	2026	Updated safety standard for heat pumps using A3 flammable refrigerants including R290; defines charge limits, leak detection requirements, and installation safety zones critical to product architecture decisions

UK F-Gas Regulation (retained EU 517/2014 as amended) and forthcoming HFC Phase-Down Schedule	regulatory_document	2026	Governs use, import, and placement on market of fluorinated refrigerants; R290 as a natural refrigerant is exempt from phase-down quotas, making it strategically advantageous for long-term product viability
BSRIA World Market for Heat Pumps 2023 Report	market_research	2026	Provides market sizing, growth forecasts, competitive landscape analysis, and technology trend data for the European and UK heat pump markets including commercial segment breakdown
Ecodesign Regulation (EU) 2016/2281 / UK Ecodesign for Energy-Related Products Regulations – Lot 1 (Space Heaters)	regulatory_standard	2026	Sets minimum seasonal energy performance requirements (SEERs, SCOP) for heat pumps placed on the UK/EU market; compliance required for product legality and mandatory energy labelling
CIBSE TM65 – Embodied Carbon in Building Services Equipment	industry_guidance	2026	Increasingly referenced by commercial building specifiers and architects for whole-life carbon assessments; R290 refrigerant choice directly reduces embodied refrigerant carbon, supporting specification arguments
Carbon Trust / EST – Electrification of Heat in Commercial Buildings Study 2022	market_research	2026	Quantifies the market opportunity for heat pumps in UK small commercial buildings, identifies barriers to adoption, and outlines cost reduction pathways relevant to the £4,500 target unit cost strategy

Data Sources

[Mixed] Constraints provided by User, Sources generated by LLM

Overall Section Grade: B

2.1 MCS 007 ^[D]

MCS Heat Pump Standard

Status: not-started

Owner: Regulatory Affairs Manager / Head of Engineering

Summary

Microgeneration Certification Scheme standard governing performance testing, documentation, installation quality, and ongoing compliance for heat pumps installed in UK buildings to qualify for government financial incentives including the Boiler Upgrade Scheme

Applicability

UK market; mandatory for BUS grant eligibility which is the primary commercial incentive driver for the target customer segment

Engineering Impact

Requires tested and declared COP/capacity data at specific rating points (including A2/W35); mandates sound power level testing; defines installer competency requirements; product data sheets must follow MCS template format

EVIDENCE REQUIRED

UKAS-accredited third-party test report to BS EN 14511 at all MCS-required rating points; sound power test report to BS EN ISO 9614; full product technical documentation; production quality audit; installer training materials

GAP ACTION

Engage MCS certification body (e.g., KIWA, BRE, NSAI) at prototype stage; commission accredited test laboratory (e.g., BSRIA, AIRMEC, or European equivalent) for performance testing; budget ~£25,000–£40,000 for initial certification

Data Sources

[Regulatory LLM] Standards extracted from context

Overall Section Grade: D

2.2 BS EN 60335-2-40 / IEC 60335-2-40:2022 ^[D]

Safety of Household and Similar Electrical Appliances – Particular Requirements for Electrical Heat Pumps, Air-Conditioners and Dehumidifiers

Status: not-started

Owner: Lead Hardware Engineer / Safety Engineer

Summary

Defines electrical and mechanical safety requirements for heat pump products, with specific annexes (GG, HH) addressing requirements for systems using flammable A3 refrigerants such as R290, including maximum charge quantities, leak detection, ventilation requirements, and ignition source controls

Applicability

UK and EU markets; required for UKCA and CE marking; particularly critical given R290 flammability classification

Engineering Impact

Maximum R290 charge in indoor unit/spaces must comply with concentration limits; outdoor unit placement rules; requirement to eliminate ignition sources in refrigerant-containing enclosures; may require integrated leak detection sensors; enclosure IP rating and cable management requirements

EVIDENCE REQUIRED

Full safety test report from UKAS/CB-scheme accredited laboratory; risk assessment for R290 charge quantity vs room volume; electrical safety test results; EMC test report; technical construction file

GAP ACTION

Conduct charge limit calculation for worst-case indoor installation scenarios in early design phase; engage safety test laboratory (e.g., TÜV, Intertek, SGS) for pre-compliance review; design ignition-source-free refrigerant compartments in outdoor unit

Data Sources

[Regulatory LLM] Standards extracted from context

Overall Section Grade: D

2.3 EN 378-1 to EN 378-4:2016+A1:2021 [D]

Refrigerating Systems and Heat Pumps – Safety and Environmental Requirements

Status: not-started

Owner: Mechanical/Refrigeration Systems Engineer

Summary

European standard series covering design, construction, testing, marking, documentation, installation, commissioning, operation, maintenance, and disposal of refrigerating systems including heat pumps; Part 2 specifically addresses construction and testing with requirements for flammable refrigerants

Applicability

UK (adopted as BS EN 378) and EU; applicable to the full refrigerant circuit design in both outdoor and indoor units; referenced by building regulations for HVAC refrigeration systems

Engineering Impact

Defines system classification based on refrigerant group (R290 = Group L2 – lower flammability) and occupancy type; mandates pressure vessel design to PED requirements; specifies relief valve sizing, leak testing pressures, and maximum allowable refrigerant quantities per occupied space; requires system pressure design to 1.1× maximum allowable working pressure

EVIDENCE REQUIRED

Design calculations demonstrating compliance with charge limits for target building occupancy class; pressure test records; material compatibility evidence; technical file with system schematic and P&ID; declaration of conformity

GAP ACTION

Classify target occupancy types at design outset; calculate maximum permissible R290 charge for split architecture; specify pressure relief valve and high-pressure cut-out set-points in line with EN 378-2; include EN 378 compliance in technical file structure from project start

Data Sources

[Regulatory LLM] Standards extracted from context

Overall Section Grade: D

2.4 SI 2016/1105 (UK) / EU 517/2014 ^[D]

F-Gas Regulation – Fluorinated Greenhouse Gases

Status: not-started

Owner: Product Compliance Manager

Summary

Regulates the use, containment, recovery, and phase-down of fluorinated greenhouse gases in refrigeration and heat pump equipment; R290 is a natural hydrocarbon refrigerant with GWP=3 and is not subject to F-Gas phase-down quotas or most operational restrictions, but product documentation must correctly identify refrigerant classification and any applicable obligations

Applicability

UK market under retained UK F-Gas Regulation; R290 is outside HFC quota system but product labelling and documentation must correctly classify the refrigerant; installation contractors working with R290 require appropriate flammable refrigerant handling competency rather than F-Gas certification

Engineering Impact

No F-Gas quota burden on R290 product; however product labelling must state refrigerant type and GWP; installer competency requirements differ from HFC systems (REFCOM Category I not strictly required but flammable refrigerant handling training mandatory); marketing opportunity to highlight GWP=3 vs R410A (GWP=2088) or R32 (GWP=675)

EVIDENCE REQUIRED

Correct refrigerant identification on product label (type, GWP, charge weight); installer guidance documentation covering safe handling of A3 flammable refrigerant; no F-Gas operator log required for R290 systems but recommended practice

GAP ACTION

Draft product label and technical datasheet with correct R290 GWP labelling; develop installer training module for R290 safe handling in commercial setting; confirm UK F-Gas regulatory position post-Brexit divergence with legal counsel

Data Sources

[Regulatory LLM] Standards extracted from context

Overall Section Grade: D

2.5 SI 2021/745 / EU 2016/2281 ^[D]

Ecodesign for Energy-Related Products – Space Heaters (Lot 1)

Status: not-started

Owner: Regulatory Affairs Manager / Test Engineer

Summary

Sets mandatory minimum energy efficiency requirements and information requirements for space heaters including heat pumps placed on the UK and EU markets; requires minimum seasonal coefficient of performance (SCOP) thresholds and mandates specific product information sheets and energy labels

Applicability

UK (UK Ecodesign Regulations SI 2021/745) and EU (Regulation 2016/2281); applies to heat pumps with rated heat output $\geq 400\text{kW}$; energy labelling (A+++ to G scale) required at point of sale

Engineering Impact

Product must achieve minimum SCOP of 3.10 in average climate (Strasbourg reference) for water-source heat output delivery; design must meet standby power, off-mode power, and noise limits; product information sheet must include declared capacity and efficiency at multiple temperature points per Annex III of the regulation

EVIDENCE REQUIRED

SCOP calculation using EN 14825 seasonal methodology; test data at part-load conditions (EN 14511 / EN 14825); completed product fiche data sheet per regulation template; energy label artwork; technical documentation file retained for 10 years

GAP ACTION

Commission EN 14825 seasonal performance testing alongside EN 14511 rating point tests to generate SCOP values; calculate energy label class; prepare product fiche and register on EPREL (European Product Registry for Energy Labelling) and UK equivalent; budget testing as part of MCS test programme to minimise duplication

Data Sources

[Regulatory LLM] Standards extracted from context

Overall Section Grade: D

3. Sizing & Spatial Optimisation [D]

INFEASIBLE SPATIAL ALLOCATION

Modules exceed available envelope.

System Envelope

Internal Dimensions (W×D×H)	5000 × 5000 × 3000 mm
Internal Floor Area	25 m²
Internal Volume	75 m³
Rules Domain	hvac_heat_pumps

Module Allocation Zone Table

Module	L × W × H (mm)	Area (m²)	Mount
compressor_assembly	2236 × 2236 × 2400	5	floor
air_source_evaporator	2236 × 2236 × 2400	5	floor
fan_air_movement	2236 × 2236 × 2400	5	floor
refrigerant_hydraulic_condenser	2236 × 2236 × 2400	5	floor
refrigerant_management	2236 × 2236 × 2400	5	floor
outdoor_chassis_enclosure	2236 × 2236 × 2400	5	floor
outdoor_control_panel	2236 × 2236 × 2400	5	floor
indoor_circulation_pump	2236 × 2236 × 2400	5	floor
indoor_safety_hydraulics	2236 × 2236 × 2400	5	floor
indoor_control_bms	2236 × 2236 × 2400	5	floor

Conflicts

- Total module footprint (50.0m²) exceeds envelope budget (21.3m²)

Data Sources

[Deterministic Solver] 3D Box Packing Constraints Engine

Overall Section Grade: B

4.1 R290 Inverter Compressor Module [D]

Maturity: draft

Purpose

Compresses low-pressure R290 vapor into high-temperature, high-pressure gas to drive the thermodynamic heating cycle.

Why It Matters

Without it, the refrigerant cannot reach the condensation temperature required to heat the hydronic circuit, resulting in complete failure to deliver thermal energy.

Technical Description

The engineered core of the heat pump cycle, utilizing an inverter-driven hermetic scroll or twin-rotary compressor highly optimized for R290 (propane) thermodynamic properties. The compressor achieves the necessary 30kW thermal output while operating at safely managed low evaporating temperatures characteristic of UK winters.

Constructed with high-grade hardened steel scroll sets to withstand the specific solubility characteristics of propane and POE (polyolester) lubricants. The integrated inverter drive rapidly modulates electrical frequency between 30Hz and 120Hz, ensuring precise load matching across the 10kW to 30kW thermal range and maximizing the Seasonal Coefficient of Performance (SCOP) at part-load conditions.

To address acoustic limitations in commercial settings (such as MCS 020 permitted development limits), the entire compressor operates on tuned elastomeric vibration isolators and is rigidly encapsulated within a heavy multi-layer acoustic jacket made of mass-loaded vinyl and open-cell flame-retardant melamine foam.

Bill of Materials (BOM) [D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

4.2 Outdoor Air Evaporator Coil [D]

Maturity: draft

Purpose

Absorbs thermal energy from the outdoor ambient air by boiling low-pressure, low-temperature liquid R290 into vapor.

Why It Matters

This is the primary heat absorption surface; if fouled or critically iced, the compressor will trip on low pressure and heating capacity drops to zero.

Technical Description

The air-source evaporator acts as the primary thermal interface with the environment, optimized for the high volume of airflow required to extract 30kW of heat from sub-zero ambient conditions. The coil is arranged in an L-shape or U-shape to maximize finface area within the constrained footprint of the outdoor unit, allowing for a low approach temperature difference (T) driving high system COP.

Fabricated using inner-grooved 7mm or 9mm outer-diameter copper tubing for enhanced internal turbulent refrigerant flow. The tubes are mechanically expanded into undulating aluminum fins treated with a hydrophilic epoxy coating. This specific coating reduces the surface tension of condensate water, accelerating drainage and delaying the onset of detrimental ice bridging across the fins.

Positioned immediately above a heated, sloped drainage pan, the module integrates structural stiffeners to withstand wind loads while minimizing harmonic resonance from the adjacent axial fans. A strategically placed thermistor provides real-time frost thickness proxy data to trigger reverse-cycle defrosts.

Bill of Materials (BOM) [D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

4.3 EC Axial Fan Array [D]

Maturity: draft

Purpose

Maintains required volumetric airflow across the evaporator coil to ensure continuous and efficient heat transfer.

Why It Matters

No airflow eliminates heat transfer at the evaporator, causing the refrigerant pressure to instantly collapse and the safety interlocks to shut down the unit.

Technical Description

Twin electronically commutated (EC) axial fans spanning 630mm to 800mm in diameter, engineered to deliver the massive volumetric flow rate required for a 30kW thermal capacity without violating strict commercial acoustic limits. EC motor technology is critical here, inherently combining high aerodynamic efficiency with deeply variable speed capability, directly reducing auxiliary power consumption and boosting the reported seasonal COP under EN 14825.

The fan blades employ a swept, biomimetic owl-wing design cast in glass-reinforced composite. This serrated trailing edge design specifically breaks down turbulent vortices, drastically reducing broad-band acoustic noise. The fans are mounted in deep, aerodynamically optimized polymer shrouds that convert dynamic pressure into static pressure, pushing air smoothly over the coil.

Controlled directly by the main outdoor PCB via 0-10V or PWM signals, the fan array modulates its RPM dynamically based on ambient temperature, compressor load, and refrigerant condensing pressure. Special anti-vibration mountings prevent motor harmonics from transmitting into the sheet metal chassis.

Bill of Materials (BOM) [D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

4.4 Brazed Plate Condenser [D]

Maturity: draft

Purpose

Transfers thermal energy from the extreme heat of the high-pressure R290 gas directly into the building's hydronic water circuit.

Why It Matters

This component is the critical thermal bridge between the flammable outdoor refrigeration cycle and the safe indoor water network; a failure here cripples heating and poses severe contamination risks.

Technical Description

The brazed plate heat exchanger (BPHE) functions as the system condenser, designed to handle the ~28-bar operating pressures of R290 while providing an enormous internal surface area for heat exchange. Engineered using a true counter-flow arrangement, high-temperature refrigerant vapor enters the top, condensing into liquid as it transfers heat to the counter-flowing water, which captures the superheated gas energy to achieve flow temperatures up to 65°C.

Stamped from high-grade 316L stainless steel plates, the unit is constructed via vacuum-brazing using pure copper as the bonding agent. The specific plate pattern (chevron angle) is mathematically optimized to induce highly turbulent water flow, maximizing thermal transmission (U-value) while minimizing pressure drop to keep circulation pump energy consumption minimal.

To prevent standby heat loss when the compressor is dormant, the entire exchanger is fully encapsulated in a precisely molded Expanded Polypropylene (EPP) thermal jacket. The module is strictly classified under the Pressure Equipment Directive (PED 2014/68/EU), necessitating stringent metallurgical validation and factory pressure testing before installation.

Bill of Materials (BOM) [D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

4.5 Refrigerant Control Array [D]

Maturity: draft

Purpose

Precise regulation of refrigerant flow, pressure, state, and cycle direction to maintain optimal superheat and enable reverse-cycle defrost.

Why It Matters

Incorrect metering causes liquid refrigerant to slug and destroy the compressor, or starves the evaporator, wrecking the COP and capacity.

Technical Description

The refrigerant management module tightly regulates the state and direction of the highly flammable R290 throughout the entirety of the closed thermodynamic cycle. The absolute core of this module is the hermetically sealed Electronic Expansion Valve (EEV). Driven by a bipolar stepper motor, the EEV modulates in micro-increments to expand high-pressure liquid refrigerant into a low-pressure mist while constantly trimming system superheat to within 1-2 Kelvin, optimizing evaporator efficiency.

Coupled with the EEV is a heavy-duty 4-way reversing valve assembly. This valve, operated by an electrical solenoid directing pilot pressure, switches the flow geometry from heating mode to reverse-cycle defrost mode, redirecting hot gas to the outdoor evaporator to clear ice build-up rapidly. The module also features a solid-core filter drier and high-resolution pressure transducers to feed exact barometric data back to the primary PCB.

To meet the stringent IEC 60335-2-40 standards for A3 flammable refrigerants, all electromechanical solenoids and coils within this module are either encapsulated to eliminate sparks or positioned strictly outside established localized leak ventilation traps. Piping structures are continuously brazed under dry nitrogen preventing internal oxidation.

Bill of Materials (BOM) [D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

4.6 Outdoor Structural Chassis [D]

Maturity: draft

Purpose

Provides the structural backbone, weatherproofing, and acoustic isolation for the outdoor mechanical components while strictly managing R290 leak ventilation paths.

Why It Matters

Failure compromises safety; inadequate ventilation architecture will cause explosive R290 pooling in the event of a leak, violating EN 378 and IEC 60335-2-40.

Technical Description

The outdoor chassis forms the physical structure and primary barrier protecting the £4,500 investment from external environments while containing sound. Fabricated from precision-folded 1.5mm and 2.0mm thick DX51D galvanized steel sheets, the exterior panels are pre-treated and finished with a C4-class marine-grade polyester powder coating to prevent corrosion over a 15-year commercial lifecycle. Access panels are secured with captive quarter-turn fasteners to allow rapid servicing by M&E contractors.

Crucially, the base tray and compressor compartment are designed with specialized gravity-drain ventilation louvers. Because R290 (propane) is significantly heavier than air, any leak will sink. The chassis topology eliminates sub-floor basins or enclosed pockets where the flammable A3 gas could accumulate near potential ignition sources, guaranteeing compliance with EN 378 flammable limit mandates.

Acoustic management is heavily integrated. The interior surfaces of the compressor chamber are lined with dual-density mass-loaded acoustic metamaterials, isolating low-frequency compressor hum, while specialized internal baffles arrest high-pitch inverter whine, critical for passing the sound power tests required for UK MCS certification.

Bill of Materials (BOM) [D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

4.7 Primary Outdoor Control & Inverter Panel [D]

Maturity: draft

Purpose

Executes localized sensory interpretation, high-voltage power manipulation, thermodynamics logic, and mission-critical ignition-safety interlocks.

Why It Matters

The entire system shuts down instantly without this brain; exposed electrical arcing here in the presence of an R290 leak guarantees deflagration.

Technical Description

The main outdoor control panel acts as the highly protected, automated brain of the refrigeration cycle. Because the unit targets a commercial 30kW output, the module handles a 3-phase, 400V electrical supply, cleanly managing up to 25 amps per phase. It houses the customized Main Control Unit (MCU) PCB, the high-power IGBT inverter drive for the compressor, filter chokes to satisfy EMC regulations, and the EEV driver circuitry.

Housed in a dedicated IP55 die-cast or fire-retardant ABS sub-enclosure positioned at the physical top of the chassis. Since R290 propane sinks, locating the electronics high and physically partitioning them from the refrigerant components practically eliminates the risk of sparking into a gas cloud. Additionally, any relays that must sit closer to the refrigerant circuit are strictly sealed solid-state, spark-proof EX/ATEX rated components.

The logic board communicates with the indoor module via a shielded, weather-resistant RS485 data link. It continuously polls over a dozen sensors (high pressure, discharge temperature, ambient air, etc.) utilizing proprietary anti-frost and capacity-mapping algorithms to keep the heat pump operating deep within its safe, highly-efficient envelope.

Bill of Materials (BOM) [D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

4.8 Indoor Hydronic Circulation Pump [D]

Maturity: draft

Purpose

Drives the primary thermal transfer fluid (water/glycol) between the outdoor condenser and the indoor distribution system at strict, continuous velocities.

Why It Matters

No circulation flow will instantly collapse heat transfer at the outdoor unit, causing a violent high-pressure trip in the refrigerant and failing to heat the building.

Technical Description

Located safely inside the commercial property within the indoor hydraulic module, the circulation pump module is the mechanical heart of the water circuit. It is sized to flawlessly drive ~1.43 L/s (5.1 m³/h) of water at a 5 Kelvin T, pushing through the resistance of the main buffer vessel, interconnecting pipework, and the outdoor brazed plate heat exchanger.

The module utilizes a high-efficiency standalone or integrated ECM (Electronically Commutated Motor) circulator pump, featuring a robust cast-iron or bronze volute designed for continuous, multi-year commercial operation. Operating on a variable speed curve, the pump adapts to changing hydrodynamic resistance from the building's zones, managed directly via PWM control from the indoor BMS controller.

Integrated directly adjacent to the pump housing are critical sensing and isolation components. A highly reliable mechanical flow switch (or ultrasonic flow sensor) provides a hard physical interlock ensuring water is moving before the outdoor compressor is permitted to fire. Dual heavy-duty DN32 (1¼") or DN40 brass isolation valves allow entirely localized pump replacement without draining the 30kW building loop.

Bill of Materials (BOM) [D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

4.9 Safety Hydraulics & Expansion Array [D]

Maturity: draft

Purpose

Manages hydronic circuit pressure, accommodates fluid thermal expansion, forcefully removes debris, and safeguards the building against pressure bursts.

Why It Matters

Protects the entire indoor water network from expanding heat-cycle pressures that would otherwise burst pipework and cause massive internal water damage.

Technical Description

The indoor safety hydraulics module is a comprehensive fluid management assembly pre-plumbed into the indoor unit to guarantee protection of the building's wet systems and to drastically reduce M&E contractor installation time. Central to this module is a high-grade brazed copper or cast brass manifold, designed to consolidate multiple safety functions into a tight physical envelope.

Mounted on this manifold is a PED-compliant 3-bar safety pressure relief valve (PRV), complete with a localized pressure gauge. An integrated connection leads to a nitrogen-charged diaphragm expansion vessel (typically 12L to 24L, though for a 30kW commercial system, this connects to larger external commercial vessels). This vessel absorbs the volumetric expansion of water as the heat pump pushes flow temperatures from 15°C ambient up to 60°C.

To ensure system longevity, especially crucial when retrofitting older legacy systems into low-carbon operations, a high-capacity combined magnetic dirt and micro-bubble air separator is factory-integrated. This component strips microscopic ferromagnetic debris and dissolved oxygen out of the return leg, definitively preventing scaling in the outdoor plate heat exchanger and stopping rust propagation.

Bill of Materials (BOM) [D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

4.10 BMS Gateway & System Controller ^[D]

Maturity: draft

Purpose

Provides macro system logic, weather compensation scaling, and standardized communication interfaces for both end-users and complex commercial Building Management Systems.

Why It Matters

Without intelligent load-matching and BMS integration, the heat pump runs blindly, destroying efficiency, missing ErP Ecodesign requirements, and precluding commercial specification.

Technical Description

The indoor control and BMS gateway serves as the primary master intelligence for the entire system, acting as the bridge between ambient meteorological conditions, building heat emitters, and the sealed outdoor R290 unit. Powered by an industrial-grade microprocessor running a Real-Time Operating System (RTOS), the controller continuously processes EN 14825 mandated weather compensation algorithms. It dynamically alters the flow temperature setpoint based on the real-time outdoor thermistor data, brutally optimizing the SCOP.

The system features native, out-of-the-box integration for commercial specification, containing a dedicated field network gateway supporting standard Modbus RTU, Modbus TCP, and BACnet IP protocols. This allows seamless daisy-chaining and integration with established commercial controllers (e.g., Trend, Siemens) mandated by property developers targeting EPC-B ratings. It also provides the interlocks for necessary domestic hot water (DHW) diverter valves, legionella cycle timers, and anti-legionella immersion heaters.

From a client interaction perspective, the front of the indoor chassis features a capacitive touchscreen Human Machine Interface (HMI). It delivers highly legible data arrays required for installer commissioning, fault diagnostics, and rapid MCS-007 performance reporting, avoiding the need for laptops simply to bleed the system or view pressure errors.

Bill of Materials (BOM) ^[D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

5. Cost Waterfall & Economics [D]

Unit Cost Breakdown

Module	Subtotal
Total Unit BOM Cost	Not computed

Non-Recurring Engineering (NRE)

Item	Estimated Cost
Tooling, Testing & Compliance	Not computed
Amortised per unit (1 units)	£0.00

Cost Reduction Paths

Strategy	Est. Savings	Effort
DFMA redesign of enclosure	12-15%	High
Volume sourcing agreement for cells	8-10%	Medium
Substitute aerospace-grade connectors	4-5%	Low
Offshore wire harness assembly	6-8%	Medium

Data Sources

[Deterministic] Cost compilation arithmetic

[LLM Estimator] Part estimates where price missing

Overall Section Grade: C

6. Failure Modes & Effects Analysis (FMEA) ^[D]

Module	Hazard	Cause	Effect	S×L=RPN	Mitigation & Verification Test
R290 Inverter Compressor Module	R290 leak pooling inside compressor jacket	Not computed	Not computed	5×2=10	Ventilated jacket base and ATEX-rated internal terminals
R290 Inverter Compressor Module	High starting current tripping local DNO grid compliance	Not computed	Not computed	4×1=4	Soft-start inverter parameterization limiting peak inrush to 1.5x running current
R290 Inverter Compressor Module	Excessive vibration transmission to chassis	Not computed	Not computed	3×3=9	FEA-designed isolation brackets and flexible copper discharge lines
Outdoor Air Evaporator Coil	Ice expansion crushing lower copper tubes	Not computed	Not computed	4×3=12	Elevated bottom coil row and dual PTC trace heaters in pan
Outdoor Air Evaporator Coil	Refrigerant maldistribution feeding liquid block back to compressor	Not computed	Not computed	4×2=8	CFD-optimized distributor sizing and electronic expansion valve tuning
Outdoor Air Evaporator Coil	Cost overrun on raw copper materials	Not computed	Not computed	3×4=12	Design for thinner-wall inner-grooved copper and secure long-term commodity pricing
EC Axial Fan Array	Resonant frequency coupling with chassis at specific RPMs	Not computed	Not computed	3×3=9	Skip-band frequency programming in the fan control logic
EC Axial Fan Array	Fingers passing through grille causing injury	Not computed	Not computed	4×1=4	Conform to strict ISO safety grille spacing standards (<10mm gaps where reachable)

EC Axial Fan Array	Ice build-up on fan blades causing imbalance	Not computed	Not computed	3×2=6	Active RPM shaking sequence during defrost and blade flex design
Brazed Plate Condenser	Water side freeze rupture bleeding water into refrigeration loop	Not computed	Not computed	5×2=10	Strict low-flow switch interlocks and automated frost-protection pump cycling
Brazed Plate Condenser	PED compliance failure holding up CE/UKCA marking	Not computed	Not computed	5×1=5	Source exclusively from Tier-1 PED-certified vendors (e.g., SWEP, Alfa Laval)
Brazed Plate Condenser	Excessive water-side pressure drop starving local heat emitters	Not computed	Not computed	3×2=6	Oversize standard plate port holes (DN32 minimum)
Refrigerant Control Array	Brazing failure causing rapid R290 leak	Not computed	Not computed	5×2=10	Automated orbital brazing on sub-assembly and 100% helium sniff testing
Refrigerant Control Array	Liquid slugging during sudden load reduction	Not computed	Not computed	4×2=8	Aggressive EEV closure algorithms layered with suction accumulator mass
Refrigerant Control Array	Transducer drift causing false low-pressure trips	Not computed	Not computed	3×3=9	Specify high-MTBF automotive-grade MEMS pressure sensors
Outdoor Structural Chassis	Ignition of pooled R290 by static or external spark	Not computed	Not computed	5×1=5	Design utterly flat, vented basepan; mandatory ground-bonding of all panels
Outdoor Structural Chassis	Panel rust compromising structural integrity and client aesthetics	Not computed	Not computed	3×3=9	Implement C4 level shot-blast, epoxy primer, and powder coat standard
Outdoor Structural Chassis	Sharp edges causing injury to installation contractors	Not computed	Not computed	3×2=6	Hemmed folds on all internal and external panel boundaries
Primary Outdoor Control & Inverter Panel	Relay spark igniting an R290 leak	Not computed	Not computed	5×1=5	Mandate solid-state relays or EX encapsulations per IEC 60335-2-40 Annex GG
Primary Outdoor Control & Inverter Panel	Electromagnetic interference disrupting local commercial networks	Not computed	Not computed	3×3=9	Integrated 3-phase line filters and shielded compartmentalization

Primary Outdoor Control & Inverter Panel	Grid voltage surge destroying the inverter drive	Not computed	Not computed	4×2=8	Heavy-duty MOVs (Metal Oxide Varistors) and over/under-voltage tripping circuits
Indoor Hydronic Circulation Pump	Pump dead-heading against closed building valves	Not computed	Not computed	3×4=12	Auto-regulating pressure mode on ECM and mandatory indoor bypass valve
Indoor Hydronic Circulation Pump	Flow switch failure leading to condenser freezing	Not computed	Not computed	5×2=10	Dual verification logic using flow switch AND delta-T sensing across the condenser
Indoor Hydronic Circulation Pump	Leakage at pump unions damaging indoor electronics	Not computed	Not computed	3×3=9	Physical internal drip tray routing to external condensate line
Safety Hydraulics & Expansion Array	Catastrophic over-pressurization of building pipework	Not computed	Not computed	5×1=5	Use only heavily certified, CE-marked safety relief valves direct from Tier 1 vendor
Safety Hydraulics & Expansion Array	Debris fouling the outdoor condenser	Not computed	Not computed	4×3=12	Mandate high-gauss internal magnetic separator on the return pipe prior to outdoor run
Safety Hydraulics & Expansion Array	Installer bypasses safety valves causing compliance failure	Not computed	Not computed	4×2=8	Mechanically swage the safety assembly into the main PCB interlock housing
BMS Gateway & System Controller	Cybersecurity vulnerability on the BACnet/IP port	Not computed	Not computed	4×2=8	Isolate the network stack and require physical local token key for remote firmware flashes
BMS Gateway & System Controller	Failure to achieve Ecodesign SCOP 3.10 via poor logic	Not computed	Not computed	5×2=10	Aggressive weather compensation curves mandated as factory default un-bypassable settings
BMS Gateway & System Controller	Loss of comms to the outdoor R290 unit	Not computed	Not computed	4×3=12	Failsafe shutdown algorithm on the outdoor unit if polling ceases for 30 seconds

Data Sources

[LLM Synthesis] FMEA Matrix Generation

Overall Section Grade: E

7. Source Attribution & Section Grading

Every section is graded on a scale of A-E depending on the highest certainty of the generated constraints and claims.

Section	Grade	Primary Source Type
1. Brief & Requirements	B	User Input & Verified Search
2. Regulatory Posture	D	LLM Knowledge Base
3. Sizing Optimization	B	Deterministic Solver
4. System Modules & Specs	D	Algorithmic Extraction + LLM
5. Economics Waterfall	C	Arithmetic Model + Estimations
6. FMEA Risk	E	Pure LLM Synthesis

8. Audit Log

Pipeline execution trace for traceability and verification.

#	Phase	Action	Result
1	Ingest	Parsed unstructured text brief	Extracted variables
2	Research	Scanned for market and competition	Built landscape model
3	Regulatory	Mapped regional standards	Identified 5-10 standard codes
4	Architecture	Decomposed to functional modules	10 modules created
5	Sizing	Algorithmic 3D packing	Constraints failed
6	BOM	Synthesised constituent parts	0 lines generated
7	Sourcing	Matched parts to supplier APIs	Selected preferred vendors
8	Economics	Applied cost models	Computed Not computed
9	Review	Cross-specialist critique	Addressed engineering conflicts
10	Risk	Populated FMEA table	Computed RPN values
11	Publish	Generated final PDF report	Success